Table 1: Effect of the PET-preservation and rigidity losses and of where they act on the test set (CT Dice: 28 patients, 38 pairs; metabolic tumor volume [MTV] and total lesion glycolysis [TLG]: 26 patients, 30 pairs); numerical values from Fig. **??**. Left: network trained with the losses, without instance optimization (IO); middle: the same network followed by IO with the same losses; right: the network trained without losses followed by IO with the losses. Values are mean±SD over patients after averaging each patient's eligible pairs. $^{\dagger}/^{\ddagger}$: significantly better/worse than without losses in the same block (for the right block: the network without losses with IO); $^{*}$: significantly different from the same losses in training and IO (two-sided Wilcoxon signed-rank tests on patient values, Holm-corrected over three comparisons within each column, $p < 0.05$).

| | Losses only in training (no IO) | | | Losses in training and IO | | | Losses only in IO | | |
|---|---|---|---|---|---|---|---|---|---|
| Losses | Dice (%) ↑ | \|ΔMTV\| (%) ↓ | \|ΔTLG\| (%) ↓ | Dice (%) ↑ | \|ΔMTV\| (%) ↓ | \|ΔTLG\| (%) ↓ | Dice (%) ↑ | \|ΔMTV\| (%) ↓ | \|ΔTLG\| (%) ↓ |
| None | 75.5±4.7 | 10.3±9.5 | 12.5±11.1 | 80.2±4.2 | 11.3±11.9 | 14.5±14.1 | – | – | – |
| PET | 73.6±4.6$^{\ddagger}$ | 8.2±9.1$^{\dagger}$ | 9.8±10.1 | 77.7±4.8$^{\ddagger}$ | 8.2±10.1 | 10.3±10.2 | 78.6±4.4$^{\ddagger *}$ | 9.3±11.6 | 11.8±11.7 |
| Rigidity | 75.5±4.7 | 8.9±8.4$^{\dagger}$ | 11.0±10.3 | 79.6±4.2$^{\ddagger}$ | 9.2±10.2 | 10.6±13.4$^{\dagger}$ | 79.6±4.2$^{\ddagger}$ | 10.0±11.3 | 11.5±14.2$^{\dagger}$ |
| PET + rigidity | 73.6±4.6$^{\ddagger}$ | 8.0±9.0$^{\dagger}$ | 9.8±10.9 | 77.3±4.9$^{\ddagger}$ | 7.9±9.7 | 10.3±10.4 | 78.3±4.4$^{\ddagger *}$ | 8.7±10.7 | 11.3±11.2 |